

Unifying the general- ξ, η leading theorem to every multiplicity

$$m \geq 1$$

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Abstract

The second GPT-6 Astra consult ([tide-log/gpt6_bigpicture_v2.md](#)) identified the $m = 1$ Lipschitz hypothesis of Programme A as a proof-architecture artefact: the freezing argument extends to a one-coordinate block, because the inserted block has exponent $q = (h + 2)/k > p$, so $c^p Z_{h+1}(c)$ is bounded at every scale and tends to zero. With that strip lemma the freezing limit, the leading theorem, its box form, the free energy and the arbitrary-order theorem are now stated uniformly for blocks of $d_0 + 1 \geq 1$ coordinates (multiplicity $M + 1 \geq 1$, logarithmic degree d_0 , resp. M), under joint continuity only. Zero ~~sorry~~/axiom.

1 The things formalised

```
theorem constStateIntegral_shift_tendsto_zero ... (k h : Fin 1 → ) ... :
  Tendsto (fun N : => N ^ p * constStateIntegral a b N k (shiftExp h 0) k' h')
    atTop ( 0)
theorem constStateIntegral_shift_tendsto_zero' ... (k h : Fin (d + 1) → ) ...
  (i : Fin (d + 1)) :
  Tendsto (fun N : => N ^ p / Real.log N ^ d
    * constStateIntegral a b N k (shiftExp h i) k' h') atTop ( 0)
-- generalised (d + 2 d + 1, M + 2 M + 1):
theorem blockStateIntegral_sub_frozen_tendsto, blockStateIntegral_tendsto,
  blockStateIntegral_isEquivalent, blockStateIntegral_sqrt_isEquivalent,
  boxIntegralGen_mult_isEquivalent, boxIntegralGen_mult_freeEnergy,
  boxIntegralGen_isEquivalent_general :
  ... (hM : (Finset.univ.filter fun i => finExp K H i = p).card = M + 1) ... :
  C : , 0 < C (fun n : => boxIntegralGen b (Real.sqrt n) K H ) ~[atTop]
  fun n : => C * (n ^ (-(p / 2)) * Real.log n ^ M)
```

2 Proof strategy

For the one-coordinate strip, two bounds: $c^q Z(c) \leq E$ for all $c > 0$ (the block envelope with $d = 0$) and $Z(c) \leq e^{\beta a^2/2} \int_0^b u^{h+1}$ for all c ; hence $c^p Z(c) \leq C_1 + E$ (cases $c \leq 1$, $c \geq 1$) and $c^p Z(c) \leq E c^{p-q} \rightarrow 0$. Dominated convergence over the noncritical box with dominator $(C_1 + E) v^{h'} (v^{k'})^{-p}$ finishes, the measurability in v coming from `StronglyMeasurable.integral_prod_right'`. The unified lemma is a case split on d_0 .

3 Roadblocks and resolutions

- `toNatFun h 0 0` with a numeral index is not syntactically `toNatFun_coe`'s $\uparrow(0 : \text{Fin } 1)$; prove the evaluation by `simp [toNatFun]`.
- `Fin (0 + 1)` has no `Subsingleton` instance found; use `Fin.ext` with `omega`.
- Generalising in place: `lake env lean` on a dependent file uses stale `.olean`s of edited imports; run the full `lake build` to see the true state.

4 Outcome

The leading term of `thm:TaylorTree` for general continuous ξ, η is now one theorem family covering every multiplicity and coordinate order. Next (Astra sequence): the finite density-transfer theorem, the parameterised Taylor kernel lemma, and the $d = 2$ anchored-subtraction milestone $Z(n) = n^{-p/2}(\frac{A}{2} \log n + B) + O(n^{-(p+\delta)/2}(1 + \log n))$.